Mikhail Razakov

C# / .NET · RUNTIME, COMPILERS & PERFORMANCE

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PROFILE

I work on CLR/CIL, compiler infrastructure, LLVM and static analysis. Current focus: SharpChecker symbolic computation at ISP RAS; previously LLVM 22 at MCST. I also build backend systems, with approximately one year of aggregate commercial work across a mixed stack.

TECHNICAL SKILLS

- C#/.NET, CIL, Reflection.Emit, DynamicMethod, CLR internals, Roslyn, BenchmarkDotNet
- LLVM 22/C++23, CFG/SSA, x86-64/SysV, Rust, differential testing; ASP.NET Core, PostgreSQL, Git/Linux

EXPERIENCE

ISP RAS · SharpChecker Oct 2026 — present

Symbolic computation and structured C# code/declaration metadata for static analysis and AI-assisted defect recognition.

MCST · compiler engineering internship Jul — Aug 2026

C++23/LLVM 22: experimental LICM with conservative legality checks; Python differential checks of before/after behavior.

Commercial development · C#/.NET & adjacent technologies approx. 1 year total

Application and backend work with C#/.NET, Unity, ASP.NET Core, SQL/PostgreSQL, file I/O and servers. Duration is candidate-reported.

PROJECTS

UniversalToolchain · modular compiler/runtime

Typed contracts and dependencies → validated LanguagePlan → runtime; Bytecode/AIR, interpreter and CIL backends. Repository manifest lists 1,324 tests; parity has dedicated scenarios.

Quark2 · generated-code invocation

DynamicMethod/Reflection.Emit, RuntimeMethodHandle and native pointers. BenchmarkDotNet compares reflection Invoke, delegates and pointer calls; CLR internals are experimental only.

x86-64 codegen · Rust

SSA/CFG, liveness, register allocation, spills, phi-lowering and SysV ABI; supported subset has differential native execution versus an IR interpreter.

LLVM Global IV / LICM · C++

Interprocedural effects and legality checks for optimizations; negative regression cases prevent unsafe transformations.

EDUCATION AND RECOGNITION

HSE University — Software Engineering.

MEPhI Junior 2026, 1st-degree diploma; LangDev 2026 speaker.